

KETAN RATAN ROY

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EDUCATION

National Institute of Technology, Rourkela

Bachelor of Technology (BTech)

2021 – 2025

CGPA: 8.22

Balaji Convent Jr College, Nagpur

Class XII

2019 – 2020

Percentage: 90.62

EXPERIENCE

Dassault Systèmes | Software Engineer

JUN 2025 - Present

- Designed and delivered **scalable Python-based rule automation services** to validate complex system constraints, reducing manual verification effort and enhancing system consistency across distributed environments.
- Developed **backend business logic** in Python with domain-specific languages that enforce system invariants, improving validation correctness, lowering error rates by 30%.
- Contributed to **core C++ modules** responsible for structured data modeling and transformations, upgraded the legacy backend logic by replacing deprecated APIs, enhancing performance, reliability, and long-term system stability.
- Implemented policy-based **access control systems** for object and attribute-level permissions, and integrated security updates into Git-driven **CI/CD pipelines** to ensure consistent, version-controlled deployments across environments.

Dassault Systèmes | Software Engineer Intern

JAN - JUN 2025

- Contributed to the development of a **real-time, low-latency, NLP-driven backend system** that enabled hands-free execution of structured commands through natural language input.
- Designed a **command-processing pipeline** using modern NLP techniques, achieving **97%+** intent accuracy.
- Integrated an **EKL and Python-based API** layer to execute system actions programmatically, facilitating automated backend workflows.
- Optimized **speech-to-text** components to reduce response time by 30%, enabling low-latency, concurrent command execution.

PROJECTS

CloudCart | Cloud-Native E-Commerce on EKS

MAR 2025

- Built and deployed a production-grade **3-tier** E-commerce platform with **8 microservices and 2 databases on Amazon EKS**, leveraging Kubernetes for scalable orchestration.
- Containerized and automated deployments using Docker, Helm, and eksctl, streamlining infrastructure setup and **CI/CD workflows**.
- Configured AWS ECR, IAM OIDC, and ALB Ingress Controller for secure image handling, high availability, and zero-downtime updates.

StudyLink | Interactive Student Collaboration Platform

OCT 2024

- Led the design and deployment of StudyLink, an innovative web application, enabling students to form **study groups** of up to **50** members, post academic questions, and exchange **video solutions** to enhance collaboration and interactive learning.
- Developed a real-time drawing tool using OpenCV and MediaPipe for **hand gesture** recognition, enabling an interactive user experience.
- Integrated the **Air Canvas feature** with a **Django app** to implement the drawing tool, enabling hands-free interaction on the platform.

DriveCall AI | AI Voice Agent for Automotive Call Centers

MAR 2024

- Built a real-time AI voice agent capable of handling over 20 concurrent sessions using **LiveKit, OpenAI GPT-4, and Deepgram STT**.
- Achieved more than **95% speech detection accuracy** using Silero VAD, supporting natural turn-taking and reducing interruptions.
- Implemented end-to-end voice interaction with TTS and STT pipelines using OpenAI and Deepgram APIs, resulting in human-like voice responses with latency under 1.2 seconds.

SKILLS

Frontend	React, Next.js, TypeScript, HTML, CSS, Tailwind CSS, Material UI
Backend	Django REST Framework, REST APIs, SQL (PostgreSQL, MySQL)
Infrastructure & DevOps	Linux (Ubuntu), Docker, Kubernetes, AWS, Git, Helm, Jenkins, Ansible, eksctl, Prometheus, Terraform
Languages	C, C++, Java, Python, Go(Golang)
Core	DBMS, OOPs, Computer Networks, Operating Systems, Distributed Systems, System Design

ACHIEVEMENTS

- Won the **NorCalHacks Hackathon**, themed around social good, competing against **350 teams** worldwide.
- Runner-Up, **Major League Hacking (MLH) – DragonHacks**.
- Positioned **4.3%** worldwide in **Leetcode Weekly Contest**.